

A Look Back at Project 1

Before we get started, let's take a look back at Project 1's `removeFirst` method.

- We'll compare a student solution with my solution:

A Look Back at Project 1

```
public T removeFirst() {
    if(!isEmpty()){
        T replace=sentinel.next.item;
        sentinel.next=sentinel.next.next;
        sentinel.next.prev=sentinel;
        size--;
        return replace;
    }
    return null;
}
```

```
public T removeFirst() {
    if (size == 0) {
        return null;
    }

    Node oldFront = sentinel.next;
    Node newFront = sentinel.next.next;

    sentinel.next = newFront;
    newFront.prev = sentinel;

    size -= 1;
    return oldFront.value;
}
```

What do you like better about mine (on the right)?

Recommendations

Make your code obvious and easy to read.

- It's fine to create variables that serve only to give names to things (performance penalty is negligible or non-existent).

Use good style.

- Consistent spacing.
- Variable naming (camelCase in Java).
- Note: This will be mandatory starting a bit after the midterm.

```
public T removeFirst() {  
    if (size == 0) {  
        return null;  
    }  
  
    Node oldFront = sentinel.next;  
    Node newFront = sentinel.next.next;  
  
    sentinel.next = newFront;  
    newFront.prev = sentinel;  
  
    size -= 1;  
    return oldFront.value;  
}
```

Lecture 9 (Inheritance 2)

Iterators, Object Methods

CS61B, Fall 2025 @ UC Berkeley

Slides credit: Josh Hug



Today's Goal: ArraySet

Lecture 9, CS61B, Fall 2025

Today's Goal: ArraySet

Iteration

- The Enhanced For Loop
- iterator, next, hasNext
- iterator, next, hasNext for ArraySet
- Iterable

Object Methods

- == vs. equals
- toString
- Better toString (Bonus)
- .of (Bonus)

Today's goal: Build an implementation of a Set called ArraySet.

- Won't be implementing any interface (for now).
- Starting from basic implementation, we'll add some "industrial strength" features to the ArraySet like iteration, equality checking, and toString.

```
ArraySet<String> S = new ArraySet<>();  
S.add("Oakland");  
S.add("Toronto");  
S.add("Minneapolis");  
S.add("Oakland"); // no effect  
S.add("Taipei");  
System.out.println(S.contains("Oakland"));
```

```
s = set()  
s.add("Oakland")  
s.add("Toronto")  
s.add("Minneapolis")  
s.add("Oakland") # no effect  
s.add("Taipei")  
print("Oakland" in s)
```

```
$ java SetExample  
true  
$ python set_example.py  
True
```

Starting point: A class `ArraySet` with the following methods:

- `add(value)`: Add the value to the `ArraySet` if it is not already present.
- `contains(value)`: Checks to see if `ArraySet` contains the key.
- `size()`: Returns number of values.

For simplicity, I'll ignore resizing.

The basic functionality is quite straightforward, so I'll avoid live coding.

ArraySet (Basic Implementation)

```
public class ArraySet<T> {  
    private T[] items;  
    private int size;  
  
    public ArraySet() {  
        items = (T[]) new Object[100];  
        size = 0;  
    }  
    ...  
}
```

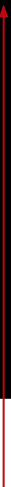
"Unchecked cast" compiler warning here.
Nothing we can do about it.

Array implementation of a Set:

- Use an array as the core data structure.
- contains(x): Checks to see if x is in the underlying array.
- add(x): Checks to see if x is in the underlying array, and if not, adds it.

ArraySet (Basic Implementation)

```
public boolean contains(T x) {  
    for (int i = 0; i < size; i += 1) {  
        if (items[i].equals(x)) {  
            return true;  
        }  
    }  
    return false;  
}
```



We used `items[i].equals(x)`, not `items[i] == x`. Recall: `==` just compares the addresses. We'll cover what `equals` does later today.

```
public void add(T x) {  
    if (!contains(x)) {  
        items[size] = x;  
        size += 1;  
    }  
}
```

Can also throw an `IllegalArgumentException` if you want to disallow null.

The Enhanced For Loop

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Today's Goal: ArraySet

Iteration

- **The Enhanced For Loop**
 - iterator, next, hasNext
 - iterator, next, hasNext for ArraySet
 - Iterable

Object Methods

- == vs. equals
- toString
- Better toString (Bonus)
- .of (Bonus)

The Enhanced For Loop

Java allows us to iterate through Lists and Sets using a convenient shorthand syntax sometimes called the “for-each loop” or “enhanced for loop”.

```
Set<Integer> javaset = new HashSet<>();  
javaset.add(5);  
javaset.add(23);  
javaset.add(42);  
for (int i : javaset) {  
    System.out.println(i);  
}
```

The Enhanced For Loop

Java allows us to iterate through Lists and Sets using a convenient shorthand syntax sometimes called the “for-each loop” or “enhanced for loop”.

- This doesn't work with our ArraySet.
- Let's strip away the magic so we can build our own classes that support this.

```
ArraySet<Integer> aset = new ArraySet<>();
aset.add(5);
aset.add(23);
aset.add(42);
for (int i : aset) {
    System.out.println(i);
}
```

```
$ javac IterationDemo
error: for-each not applicable to expression type
        for (int i : aset) {
                      ^
required: array or java.lang.Iterable
found:    ArraySet<Integer>
```

iterator, next, hasNext

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Today's Goal: ArraySet

Iteration

- The Enhanced For Loop
- **iterator, next, hasNext**
- iterator, next, hasNext for ArraySet
- Iterable

Object Methods

- == vs. equals
- toString
- Better toString (Bonus)
- .of (Bonus)

Why Doesn't the Enhanced For Loop Work?

The enhanced for loop works by first calling the `.iterator` method of the object.

- This returns an object of type `Iterator<Integer>`.
- The `Iterator` interface has its own API for fetching values one-by-one:
 - `hasNext`: Tells us whether there are more values.
 - `next`: gets the next value.

```
ArraySet<Integer> aset = new ArraySet<>();
aset.add(5);
aset.add(23);
aset.add(42);
for (int i : aset) { ← What is missing?
    System.out.println(i);
}
```

How Iteration Really Works

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.

Set.java: `public Iterator<E> iterator();`

Suppose we have a `Set<Integer>` called `javaset`.

- In that case, we can iterate with either of the two equivalent pieces of code.
- Left code is shorthand for right code, i.e. the code on the left is LITERALLY doing the thing on the right.

```
for (int x : javaset) {  
    System.out.println(x);  
}
```

“Nice” iteration.

```
Iterator<Integer> seer  
    = javaset.iterator();  
  
while (seer.hasNext()) {  
    int x = seer.next();  
    System.out.println(x);  
}
```

“Ugly” iteration.

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.

javaset:

5	23	42
---	----	----

```
Iterator<Integer> seer
    = javaset.iterator();
while (seer.hasNext()) {
    System.out.println(seer.next());
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.



javaset:

5	23	42
---	----	----

```
$ java IteratorDemo.java
```

```
→ Iterator<Integer> seer  
    = javaset.iterator();  
while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.



javaset:

5	23	42
---	----	----

```
$ java IteratorDemo.java
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
→ while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.



```
$ java IteratorDemo.java  
5
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
while (seer.hasNext()) {  
→   System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.



```
$ java IteratorDemo.java  
5
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
→ while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.

javaset:

5	23	42
---	----	----



```
$ java IteratorDemo.java  
5  
23
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.

javaset:

5	23	42
---	----	----

True



```
$ java IteratorDemo.java  
5  
23
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
→ while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.

javaset:

5	23	42
---	----	----

42



```
$ java IteratorDemo.java  
5  
23  
42
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
while (seer.hasNext()) {  
→   System.out.println(seer.next());  
}
```

How Iterators Work

An alternate, uglier way to iterate through a Set is to use the `iterator()` method.

javaset:

5	23	42
---	----	----

False



```
$ java IteratorDemo.java  
5  
23  
42
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
→ while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

iterator, next, hasNext for ArraySet

Lecture 9, CS61B, Fall 2025

Today's Goal: ArraySet

Iteration

- The Enhanced For Loop
- iterator, next, hasNext
- **iterator, next, hasNext for ArraySet**
- Iterable

Object Methods

- == vs. equals
- toString
- Better toString (Bonus)
- .of (Bonus)

The secret: The code on the left is just shorthand for the code on the right. For code on right to compile, which checks does the compiler need to do?

- A. Does the Set interface have an iterator() method?
- B. Does the Set interface have next/hasNext() methods?
- C. Does the Iterator interface have an iterator method?
- D. Does the Iterator interface have next/hasNext() methods?



```
Set<Integer> javaset = new HashSet<Integer>();
```

```
for (int x : javaset) {  
    System.out.println(x);  
}
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

The Secret of the Enhanced For Loop

The secret: The code on the left is just shorthand for the code on the right. For code on right to compile, which checks does the compiler need to do?

- A. Does the Set interface have an iterator() method?
- B. Does the Set interface have next/hasNext() methods?
- C. Does the Iterator interface have an iterator method?
- D. Does the Iterator interface have next/hasNext() methods?

```
Set<Integer> javaset = new HashSet<Integer>();
```

```
for (int x : javaset) {  
    System.out.println(x);  
}
```

```
Iterator<Integer> seer  
    = javaset.iterator();  
while (seer.hasNext()) {  
    System.out.println(seer.next());  
}
```

Supporting Ugly Iteration in ArraySets

To support ugly iteration:

- Add an `iterator()` method to `ArraySet` that returns an `Iterator<T>`.
- The `Iterator<T>` that we return should have a useful `hasNext()` and `next()` method.

`Iterator<T>`

```
public interface Iterator<T> {  
    boolean hasNext();  
    T next();  
}
```

```
Iterator<Integer> aseer  
    = aset.iterator();  
  
while (aseer.hasNext()) {  
    System.out.println(aseer.next());  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    public static void main(String[] args) {  
        ArraySet<Integer> aset = new ArraySet<>();  
        aset.add(5);  
        aset.add(23);  
        aset.add(42);  
  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    public static void main(String[] args) {  
        ArraySet<Integer> aset = new ArraySet<>();  
        aset.add(5);  
        aset.add(23);  
        aset.add(42);  
  
        Iterator<Integer> aseer = aset.iterator();  
  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    public static void main(String[] args) {  
        ArraySet<Integer> aset = new ArraySet<>();  
        aset.add(5);  
        aset.add(23);  
        aset.add(42);  
  
        Iterator<Integer> aseer = aset.iterator();  
  
        while (aseer.hasNext()) {  
  
        }  
  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    public static void main(String[] args) {  
        ArraySet<Integer> aset = new ArraySet<>();  
        aset.add(5);  
        aset.add(23);  
        aset.add(42);  
  
        Iterator<Integer> aseer = aset.iterator();  
  
        while (aseer.hasNext()) {  
            int i = aseer.next();  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    public static void main(String[] args) {  
        ArraySet<Integer> aset = new ArraySet<>();  
        aset.add(5);  
        aset.add(23);  
        aset.add(42);  
  
        Iterator<Integer> aseer = aset.iterator();  
  
        while (aseer.hasNext()) {  
            int i = aseer.next();  
            System.out.println(i);  
        }  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
  
    public Iterator<T> iterator() {  
  
    }  
  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ???;  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {
    /** returns an iterator (a.k.a. seer) into ME */
    public Iterator<T> iterator() {
        return new ???;
    }

    private class ArraySetIterator implements Iterator<T> {

    }
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ???;  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
  
        public boolean hasNext() {  
  
        }  
  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ???;  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
  
        public boolean hasNext() {  
  
        }  
  
        public T next() {  
  
        }  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
  
        public boolean hasNext() {  
  
        }  
  
        public T next() {  
  
        }  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public boolean hasNext() {  
        }  
  
        public T next() {  
  
        }  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
  
        }  
  
        public boolean hasNext() {  
  
        }  
  
        public T next() {  
  
        }  
    }  
}
```

Coding Demo: Iteration

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
        }  
  
        public T next() {  
  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
            return wizPos < size;  
        }  
  
        public T next() {  
  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
            return wizPos < size;  
        }  
  
        public T next() {  
            T returnItem = items[wizPos];  
  
            }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
            return wizPos < size;  
        }  
  
        public T next() {  
            T returnItem = items[wizPos];  
            wizPos += 1;  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
            return wizPos < size;  
        }  
  
        public T next() {  
            T returnItem = items[wizPos];  
            wizPos += 1;  
            return returnItem;  
        }  
    }  
}
```

Completed ArraySet iterator Method

To support ugly iteration:

- Add an `iterator()` method to `ArraySet` that returns an `Iterator<T>`.
- The `Iterator<T>` that we return should have a useful `hasNext()` and `next()` method.

```
private class ArraySetIterator implements Iterator<T> {  
    private int wizPos;  
    public ArraySetIterator() { wizPos = 0; }  
    public boolean hasNext() { return wizPos < size; }  
    public T next() {  
        T returnItem = items[wizPos];  
        wizPos += 1;  
        return returnItem;  
    }  
}  
  
public Iterator<T> iterator() {  
    return new ArraySetIterator();  
}
```

Iterable

Lecture 9, CS61B, Fall 2025

Today's Goal: ArraySet

Iteration

- The Enhanced For Loop
- iterator, next, hasNext
- iterator, next, hasNext for ArraySet
- **Iterable**

Object Methods

- == vs. equals
- toString
- Better toString (Bonus)
- .of (Bonus)

The Enhanced For Loop

Our code now supports “ugly” iteration, but enhanced for loop still doesn’t work.

The problem: Java isn’t smart enough to realize that our `ArraySet` has an `iterator()` method.

- Luckily there’s an interface for that.

```
ArraySet<Integer> aset = new ArraySet<>();
aset.add(5);
aset.add(23);
aset.add(42);
for (int i : aset) {
    System.out.println(i);
}
```

```
$ javac IterationDemo
error: for-each not applicable to expression type
        for (int i : aset) {
                        ^
required: array or java.lang.Iterable
found:    ArraySet<Integer>
```

For-each Iteration And ArraySets

To support the enhanced for loop, we need to make `ArraySet` implement the `Iterable` interface.

- There are also some default methods in `Iterable`, not shown.

```
public interface Iterable<T> {  
    Iterator<T> iterator();  
}
```

```
public class ArraySet<T> implements Iterable<T> {  
    ...  
    public Iterator<T> iterator() { ... }  
}
```

Iterable<T>



ArraySet<T>

ArraySet.java

```
public class ArraySet<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
            return wizPos < size;  
        }  
  
        public T next() {  
            T returnItem = items[wizPos];  
            wizPos += 1;  
            return returnItem;  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    /** returns an iterator (a.k.a. seer) into ME */  
    public Iterator<T> iterator() {  
        return new ArraySetIterator();  
    }  
  
    private class ArraySetIterator implements Iterator<T> {  
        private int wizPos;  
  
        public ArraySetIterator() {  
            wizPos = 0;  
        }  
  
        public boolean hasNext() {  
            return wizPos < size;  
        }  
  
        public T next() {  
            T returnItem = items[wizPos];  
            wizPos += 1;  
            return returnItem;  
        }  
    }  
}
```

The Iterable Interface

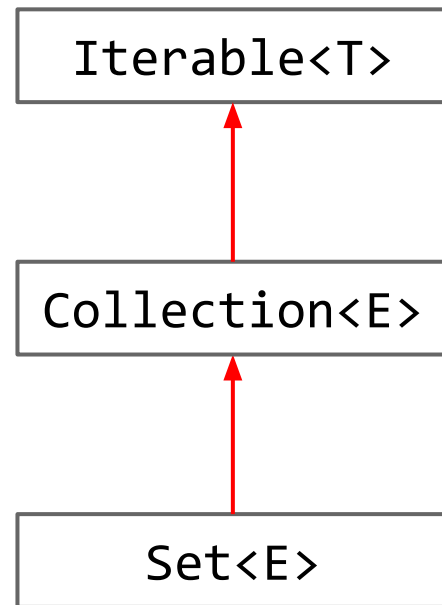
By the way, this is how Set works as well.

- Source code for Iterable: [Link](#), Set: [Link](#), Collection: [Link](#).

```
public interface Iterable<T> {  
    Iterator<T> iterator(); ...  
}
```

```
public interface Collection<E> extends Iterable<E> {  
    public Iterator<E> iterator();  
}
```

```
public interface Set<E> extends Collection<E> {  
    public Iterator<E> iterator();  
}
```



To support the enhanced for loop:

- Add an `iterator()` method to your class that returns an `Iterator<T>`.
- The `Iterator<T>` returned should have a useful `hasNext()` and `next()` method.
- Add `implements Iterable<T>` to the line defining your class.

We'll do this at the last two tasks of project 2.

Quick Note on Autoboxing/Unboxing

Note below that the next function returns an Integer.

- Yet, we are assigning `int i = aseer.next();`

```
public class ArraySet<T> {  
    public static void main(String[] args) {  
        ArraySet<Integer> aset = new ArraySet<>();  
        aset.add(5); aset.add(23); aset.add(42);  
  
        Iterator<Integer> aseer = aset.iterator();  
        while (aseer.hasNext()) {  
            int i = aseer.next();  
            System.out.println(i);  
        }  
    }  
}
```

Quick Note on Autoboxing/Unboxing

In Java, it is valid to assign Integers to ints and vice versa.

- Both lines of code below work fine.

```
Integer I = 1;  
int i = I;  
  
int iii = 3;  
Integer I = i;
```

Let's see what happens when we run the code:

- [Link](#)

Quick Note on Autoboxing/Unboxing

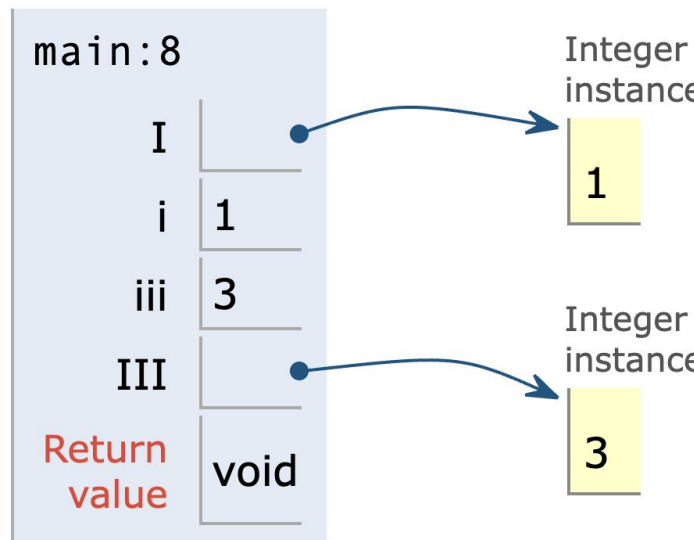
In Java, it is valid to assign Integers to ints and vice versa.

- Both lines of code below work fine.

```
Integer I = 1;  
int i = I;  
  
int iii = 3;  
Integer III = iii;
```

Let's see what happens when we run the code:

- [Link](#)



Converting an `int` to an `Integer` using `=` is called autoboxing.

- And the other direction is called unboxing.

You should 99% of the time use `int`, not `Integer`.

- The only time to use `Integer` is when you are picking a generic type, e.g. `ArraySet<Integer>`.
- Not particularly relevant in 61B, but autoboxing/unboxing slows down your code (takes time to box/unbox).

toString

Lecture 9, CS61B, Fall 2025

Today's Goal: `ArraySet`
Iteration

- The Enhanced For Loop
- `iterator`, `next`, `hasNext`
- `iterator`, `next`, `hasNext` for `ArraySet`
- `Iterable`

Object Methods

- **toString**
- `==` vs. `equals`
- Better `toString` (Bonus)
- `.of` (Bonus)

All classes are hyponyms of Object.

- **String toString()**
- `boolean equals(Object obj)`
- `int hashCode()`
- `Class<?> getClass()`
- `protected Object clone()`
- `protected void finalize()`
- `void notify()`
- `void notifyAll()`
- `void wait()`
- `void wait(long timeout)`
- `void wait(long timeout, int nanos)`

Today

Coming later.

Won't discuss or use in 61B.

toString()

The `toString()` method provides a string representation of an object.

- `System.out.println(Object x)` calls `x.toString()`
 - If you're curious: [println](#) calls [String.valueOf](#) which calls `toString`

```
Set<Integer> javaset = new HashSet<>();  
javaset.add(5);  
javaset.add(23);  
javaset.add(42);  
  
System.out.println(javaset);
```

```
$ java JavaSetPrintDemo  
[5, 23, 42]
```

toString()

The `toString()` method provides a string representation of an object.

- `System.out.println(Object x)` calls `x.toString()`
- The [implementation of toString\(\) in Object](#) is the the name of the class, then an @ sign, then the memory location of the object.
 - See 61C for what the “memory location” really means.

```
ArraySet<Integer> aset = new ArraySet<>();  
aset.add(5);  
aset.add(23);  
aset.add(42);  
  
System.out.println(aset);
```

```
$ java ArraySetPrintDemo  
ArraySet@75412c2f
```

Let's try implementing toString for ArraySet.

Coding Demo: toString

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
```

```
public String toString() {
```

}

}

Coding Demo: toString

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public String toString() {
```

```
}
```

```
}
```

Coding Demo: toString

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
    @Override
    public String toString() {
        String returnString = "{";

        }
}
```

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public String toString() {  
        String returnString = "{";  
        for (T item : this) {  
  
        }  
  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public String toString() {  
        String returnString = "{";  
        for (T item : this) {  
            returnString += item.toString();  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public String toString() {  
        String returnString = "{";  
        for (T item : this) {  
            returnString += item.toString();  
            returnString += ", ";  
        }  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public String toString() {  
        String returnString = "{";  
        for (T item : this) {  
            returnString += item.toString();  
            returnString += ", ";  
        }  
        returnString += "}";  
    }  
}
```

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public String toString() {  
        String returnString = "{";  
        for (T item : this) {  
            returnString += item.toString();  
            returnString += ", ";  
        }  
        returnString += "}";  
        return returnString;  
    }  
}
```

One approach is shown below.

- Warning: This code is slow. Intuition: Adding even a single character to a string creates an entirely new string. Will discuss why at end of course.

```
@Override
public String toString() {
    String returnString = "{";
    for (int i = 0; i < size; i += 1) {
        returnString += keys[i];
        returnString += ", ";
    }
    returnString += "}";
    return returnString;
}
```

Spoiler: It's
because
Strings are
"immutable".

keys[i] might not be a string,
but Java will automatically call
toString so that you can add
it to a string.

You can modify this code to avoid the
extra comma at the end, if you want.

Much faster approach is shown below.

- Intuition: Append operation for a StringBuilder is fast.
- See the videos for more details about StringBuilder.

```
@Override
public String toString() {
    StringBuilder returnSB = new StringBuilder("{}");
    for (int i = 0; i < size; i += 1) {
        returnSB.append(items[i]);
        returnSB.append(", ");
    }
    returnSB.append("}");
    return returnSB.toString();
}
```

== vs. equals

Lecture 9, CS61B, Fall 2025

Today's Goal: ArraySet
Iteration

- The Enhanced For Loop
- iterator, next, hasNext
- iterator, next, hasNext for ArraySet
- Iterable

Object Methods

- toString
- **== vs. equals**
- Better toString (Bonus)
- .of (Bonus)

All classes are hyponyms of Object.

- `String toString()`
- **`boolean equals(Object obj)`**
- `int hashCode()`
- `Class<?> getClass()`
- `protected Object clone()`
- `protected void finalize()`
- `void notify()`
- `void notifyAll()`
- `void wait()`
- `void wait(long timeout)`
- `void wait(long timeout, int nanos)`

Coming in another lecture soon.

Coming later.

Won't discuss or use in 61B.

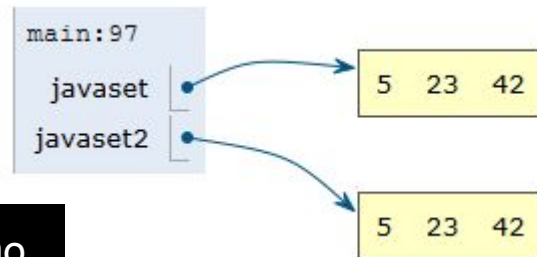
Equals vs. ==

As mentioned in an offhand manner previously, `==` and `.equals()` behave differently.

- `==` compares the bits. For references, `==` means “referencing the same object.”

```
Set<Integer> javaset = Set.of(5, 23, 42);  
Set<Integer> javaset2 = Set.of(5, 23, 42);  
System.out.println(javaset == javaset2);
```

```
$ java EqualsDemo  
False
```

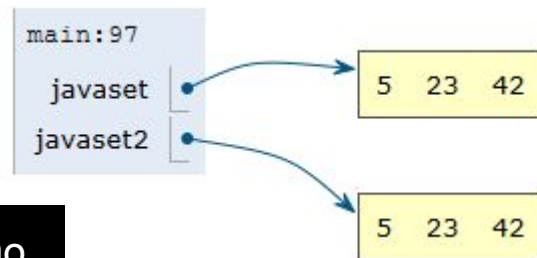


Equals vs. ==

As mentioned in an offhand manner previously, `==` and `.equals()` behave differently.

- `==` compares the bits. For references, `==` means “referencing the same object.”

```
Set<Integer> javaset = Set.of(5, 23, 42);  
Set<Integer> javaset2 = Set.of(5, 23, 42);  
System.out.println(javaset.equals(javaset2));
```



```
$ java EqualsDemo  
True
```

To test equality in the sense we usually mean it, use:

- `.equals` for classes. Requires writing a `.equals` method for your classes.
 - [Default implementation](#) of `.equals` uses `==` (probably not what you want).
- BTW: Use `Arrays.equals` or `Arrays.deepEquals` for arrays.

this is a reference to the current object. Example from lecture 2:

```
public Dog maxDog(Dog uddaDog) {  
    if (size > uddaDog.size) {  
        return this;  
    }  
    return uddaDog;  
}
```

Naturally, can also use `this` to access your own instance variables or methods.

- Unlike Python, where `self` is mandatory, using `this` is not mandatory.
- Two code snippets below are exactly identical in behavior.

```
public Dog maxDog(Dog o) {  
    if (this.size > o.size) {  
        return this;  
    }  
    return o;  
}
```

```
public Dog maxDog(Dog o) {  
    if (size > o.size) {  
        return this;  
    }  
    return o;  
}
```

Naturally, can also use `this` to access your own instance variables or methods.

- Unlike Python, where `self` is mandatory, using `this` is not mandatory.
- If there's ever a name conflict where **a local variable has the same name as an instance variable (left)**, you **must** use `this` if you want to access the instance variable.

```
public Dog(int size) {  
    size = size;  
}
```

Does nothing!

```
public Dog(int s) {  
    size = s;  
}
```

Works correctly!

```
public Dog(int size) {  
    this.size = size;  
}
```

Works correctly!

```
public Dog(int s) {  
    this.size = s;  
}
```

Works correctly!

The Default Implementation of Equals

Below, we see the actual code for the default equals method in Object.java.

- Code available here if you want to poke around:
<https://github.com/openjdk/jdk17/blob/master/src/java.base/share/classes/java/lang/Object.java#L162>

```
public class Object {  
  
    ...  
  
    public boolean equals(Object obj) {  
        return (this == obj);  
    }  
}
```

The Default Implementation of Equals

```
ArraySet<Integer> aset = new ArraySet<>();  
aset.add(5);  
aset.add(23);  
aset.add(42);
```

```
System.out.println(aset);
```

```
ArraySet<Integer> aset2 = new ArraySet<>();  
aset2.add(5);  
aset2.add(23);  
aset2.add(42);
```

```
System.out.println(aset.equals(aset2));
```

```
$ java EqualsDemo  
False
```

Returns false because
the default
implementation of equals
just uses ==.

instanceOf Demo (for hypothetical Dog equals method)

The instanceof keyword is very powerful in Java.

- Checks to see if o is pointing at a Dog.
If no, returns false.
- If yes, returns true *and* puts o in a new variable of type Dog called uddaDog.
- Works correctly, even if o is null.

```
@Override
public boolean equals(Object o) {
    if (o instanceof Dog uddaDog) {
        return this.size == uddaDog.size;
    }
    return false;
}
```

Let's try to write ArraySet's equals method.

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
```

```
public boolean equals(HashSet o) {
```

}

}

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
    @Override
    public boolean equals(Object o) {
```

}

}

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
    @Override
    public boolean equals(Object o) {
        if (o instanceof ArraySet oas) {

        }
    }
}
```

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
    @Override
    public boolean equals(Object o) {
        if (o instanceof ArraySet oas) {

        }
        // o is not an arrayset, so return false
        return false;
    }
}
```

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public boolean equals(Object o) {  
        if (o instanceof ArraySet oas) {  
            // check sets are of the same size  
            if (oas.size != this.size) {  
                return false;  
            }  
  
        }  
        // o is not an arrayset, so return false  
        return false;  
    }  
}
```

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public boolean equals(Object o) {  
        if (o instanceof ArraySet oas) {  
            // check sets are of the same size  
            if (oas.size != this.size) {  
                return false;  
            }  
  
            // check that all of MY items are in the other array set  
  
        }  
        // o is not an arrayset, so return false  
        return false;  
    }  
}
```

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public boolean equals(Object o) {  
        if (o instanceof ArraySet oas) {  
            // check sets are of the same size  
            if (oas.size != this.size) {  
                return false;  
            }  
  
            // check that all of MY items are in the other array set  
            for (T x : this) {  
  
            }  
  
        }  
        // o is not an arrayset, so return false  
        return false;  
    }  
}
```

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {  
    @Override  
    public boolean equals(Object o) {  
        if (o instanceof ArraySet oas) {  
            // check sets are of the same size  
            if (oas.size != this.size) {  
                return false;  
            }  
  
            // check that all of MY items are in the other array set  
            for (T x : this) {  
                if (!oas.contains(x)) {  
                    return false;  
                }  
            }  
        }  
        // o is not an arrayset, so return false  
        return false;  
    }  
}
```

Coding Demo: equals

ArraySet.java

```
public class ArraySet<T> implements Iterable<T> {
    @Override
    public boolean equals(Object o) {
        if (o instanceof ArraySet oas) {
            // check sets are of the same size
            if (oas.size != this.size) {
                return false;
            }

            // check that all of MY items are in the other array set
            for (T x : this) {
                if (!oas.contains(x)) {
                    return false;
                }
            }

            return true;
        }
        // o is not an arrayset, so return false
        return false;
    }
}
```

ArraySet equals

The code below is pretty close to what a standard equals method looks like.

```
@Override
public boolean equals(Object other) {
    if (this == other) { return true; }

    if (other instanceof ArraySet otherSet) {
        if (this.size != otherSet.size) { return false; }
        for (T x : this) {
            if (!otherSet.contains(x)) {
                return false;
            }
        }
        return true;
    }
    return false;
}
```

Technically a raw type
without a type placeholder
like `ArraySet<T>`, but
don't worry about it.

ArraySet equals

The code below is pretty close to what a standard equals method looks like.

```
@Override
public boolean equals(Object other) {
    if (this == other) { return true; }
    if (other instanceof ArraySet otherSet) {
        if (this.size != otherSet.size) { return false; }
        for (T x : this) {
            if (!otherSet.contains(x)) {
                return false;
            }
        }
        return true;
    }
    return false;
}
```

Doesn't affect correctness, but saves us time if this and other reference the same object.

Historical Note: Old School Equals Methods

Equals methods written before March 2021 were ugly.

- Lots of manual type checking, “casting”, and null checking.
- See the CS61B [2021 slides](#).
- You should avoid the old way (explicit casting).

```
@Override // OLD SCHOOL APPROACH. NOT PREFERRED IN 61B.
public boolean equals(Object o) {
    if (o == null) { return false; }
    if (this == o) { return true; } // optimization
    if (this.getClass() != o.getClass()) { return false; }
    ArraySet<T> other = (ArraySet<T>) o;
    ...
}
```

We built our own Array based Set implementation.

To make it more industrial strength we:

- Added an exception if a user tried to add null to the set. (See videos.)
 - There are other ways to deal with nulls. Our choice was arguably bad.
- Added support for “ugly” then “nice” iteration.
 - Ugly iteration: Creating a subclass with next and hasNext methods.
 - Nice iteration: Declaring that ArraySet implements Iterable.
- Added a toString() method.
 - Beware of String concatenation.
- Added an equals(Object) method.
 - Used instanceof to check the class of the passed object.

Better toString (Bonus)

Lecture 9, CS61B, Fall 2025

Today's Goal: ArraySet
Iteration

- The Enhanced For Loop
- iterator, next, hasNext
- iterator, next, hasNext for ArraySet
- Iterable

Object Methods

- == vs. equals
- toString
- **Better toString (Bonus)**
- .of (Bonus)

Can use the `String.join` method to convert list of strings into a single string.

```
@Override
public String toString() {
    List<String> listOfItems = new ArrayList<>();
    for (T x : this) {
        listOfItems.add(x.toString());
    }
    return "{" + String.join(", ", listOfItems) + "}";
}
```

ArraySet.of (Bonus)

Lecture 9, CS61B, Fall 2025

Today's Goal: ArraySet
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- == vs. equals
- toString
- Better toString (Bonus)
- **.of (Bonus)**

We can write our own of method as follows:

- Below stuff is a so-called “var arg”.
 - Object passed is an array.
 - Values filled out using comma separated syntax.

```
public static <Glerp> ArraySet<Glerp> of(Glerp... stuff) {  
    ArraySet<Glerp> returnSet = new ArraySet<Glerp>();  
    for (Glerp x : stuff) {  
        returnSet.add(x);  
    }  
    return returnSet;  
}
```

Attendance link: <https://www.yellkey.com/how>

To-do:

- Project 2, due September 22. You now know everything you need to know for the entire project.

Today was the last in scope lecture for the midterm!

- More on midterm studying next time.

